

Total No. of Questions : 6]

SEAT No. :

P3723

[Total No. of Pages : 2

Engg. - 52
T.E. (IT) (Semester - I)
THEORY OF COMPUTATION (In Sem.)
(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

Unit - I

- Q1)** a) Define and compare NFA and DFA [4]
b) Design finite automation for the following [6]
i) FA which reads strings made up of $\{0, 1\}$ and accepts only those strings which end in either '00' or '11'.
ii) FA which accepts only those strings with 'a' at every even position. $\Sigma = \{a, b\}$.

OR

- Q2)** a) Define and compare Moore and Mealy machines. [4]
b) Construct Mealy machine equivalent to the given Moore machine [6]

	0	1	O/P
q0	q0	q1	N
q1	q0	q2	N
q2	q0	q3	N
q3	q0	q3	Y

Start state : q0 ; Final state : q3

P.T.O.

Unit - II

Q3) a) Define regular expressions. Give RE for the following over $\Sigma = \{0,1\}$ [6]

i) All binary strings with at least one 0.

ii) All binary strings with at most one 0.

b) Convert the given left linear grammar G_L to its equivalent right linear grammar G_R . [4]

$S \rightarrow B1 / A0 / C0$

$A \rightarrow C0 / A1 / B1 / 0$

$B \rightarrow B1 / 1$

$C \rightarrow A0$

OR

Q4) a) Convert the given regular grammar to its equivalent FA. [8]

$S \rightarrow aS / bS / aA$

$A \rightarrow bB$

$B \rightarrow aC$

$C \rightarrow a$

b) Define regular grammar. [2]

Unit - III

Q5) a) Write a CFG for [6]

i) Generating strings containing any no. (zero or more) of a's. $\Sigma = \{a\}$

ii) The RE a^*b^*

b) Write a note on Chomsky hierarchy. [4]

OR

Q6) a) With examples define CNF and GNF. [4]

b) Simplify the given grammar [6]

$S \rightarrow aAa$

$A \rightarrow Sb / bCC / DaA$

$C \rightarrow abb / DD$

$E \rightarrow aC$

$D \rightarrow aDa$

